

Different types of data available

- .txt,
 - it is a text file
 - no package is required
- .csv,
 - comma serperated values stored in excel
 - package is required
 - direct csv package or pandas
- .xlsx,
 - it is a excel file
 - package is required
- .json,
 - it has key:value pairs
 - package is required
- .xml,
 - it has information attached with height width
 - package is required
- .pdf,
 - package is required
- .docx, word document
 - package is required
- .tfrecord data stored in tesnors (encoded)
 - package is required
- .delta it is encoded format (DE pyspark)
 - package is required
- .parquet it is also encoded format (DE pyspark)
 - package is required

```
In [6]: # read the text file
# file name: mbox_short.txt
# q1) what is the file location:
#         In which folder the file is existed

# Location:"C:\Users\omkar\OneDrive\Documents\Data science\Naresh IT\Datafi

# Q2) in that above location there are so many files are avialable
# file_name:"mbox-short"

# Q3) file extension: .txt

# we need to combine above three

# patterns:<Location>\<filename>.<type of file>

#"C:\Users\omkar\OneDrive\Documents\Data science\Naresh IT\Datafiles\mbox-s"
```

```
In [4]: unicode: single forward slash

give the double slash

we can read using single slash

hello
```

```
In [7]: # Step-1: make full file path
file_path="C:\\Users\\omkar\\OneDrive\\Documents\\Data science\\Naresh IT\\Datafiles\\mbox-short.txt"
file_path
```

```
Out[7]: 'C:\\Users\\omkar\\OneDrive\\Documents\\Data science\\Naresh IT\\Datafiles\\mbox-short.txt'
```

```
In [9]: # Step-2: open the file path

file=open(file_path)
file
```

```
Out[9]: <_io.TextIOWrapper name='C:\\Users\\omkar\\OneDrive\\Documents\\Data science\\Naresh IT\\Datafiles\\mbox-short.txt' mode='r' encoding='cp1252'>
```

```
In [10]: # step-3: start read
file.read()
```

```
Out[10]: 'From stephen.marquard@uct.ac.za Sat Jan  5 09:14:16 2008\nReturn-Path: <postmaster@collab.sakaiproject.org>\nReceived: from murder (mail.umich.edu [141.211.14.90])\n\tby frankenstein.mail.umich.edu (Cyrus v2.3.8) with LMTPA;\n\tSat, 05 Jan 2008 09:14:16 -0500\nX-Sieve: CMU Sieve 2.3\nReceived: from murder ([unix socket])\n\tby mail.umich.edu (Cyrus v2.2.12) with LMTPA;\n\tSat, 05 Jan 2008 09:14:16 -0500\nReceived: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu [141.211.14.79])\n\tby flawless.mail.umich.edu () with ESMTP id m05EEFR1013674;\n\tSat, 5 Jan 2008 09:14:15 -0500\nReceived: FROM paploo.uhi.ac.uk (app1.prod.collab.uhi.ac.uk [194.35.219.184])\n\tBY holes.mr.itd.umich.edu ID 477F90B0.2DB2F.12494 ;\n\t5 Jan 2008 09:14:10 -0500\nReceived: from paploo.uhi.ac.uk (localhost [127.0.0.1])\n\tby paploo.uhi.ac.uk (Postfix) with ESMTP id 5F919BC2F2;\n\tSat,  5 Jan 2008 14:10:05 +0000 (GMT)\nMessage-ID: <200801051412.m05ECIaH010327@nakamura.uits.iupui.edu>\nMime-Version: 1.0\nContent-Transfer-Encoding: 7bit\nReceived: from prod.collab.uhi.ac.uk ([194.35.219.182])\n\tby paploo.uhi.ac.uk (JAMES SMTP Server 2.1.3) with SMTP ID 899\n\tfor <source@collab.sakaiproject.org>;\n\tSat, 5 Jan 2008 14:09:50 +0000 (GMT)\nReceived: from nakamura.uits.iupui.edu (nakamura.uits.iupui.edu [134.68.220.122])\n\tby shmi.uhi.ac.uk (Postfix) with ESMTP id 4345042002;\n\tSat,  5 Jan 2008 14:09:50 +0000 (GMT)

'

```

```
In [14]: file_path="C:\\Users\\omkar\\OneDrive\\Documents\\Data science\\Naresh IT\\  
file=open(file_path)  
print(file.read())
```

```
From stephen.marquard@uct.ac.za Sat Jan  5 09:14:16 2008  
Return-Path: <postmaster@collab.sakaiproject.org>  
Received: from murder (mail.umich.edu [141.211.14.90])  
        by frankenstein.mail.umich.edu (Cyrus v2.3.8) with LMTPA;  
        Sat, 05 Jan 2008 09:14:16 -0500  
X-Sieve: CMU Sieve 2.3  
Received: from murder ([unix socket])  
        by mail.umich.edu (Cyrus v2.2.12) with LMTPA;  
        Sat, 05 Jan 2008 09:14:16 -0500  
Received: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu [141.211.1  
4.79])  
        by flawless.mail.umich.edu () with ESMTP id m05EEFR1013674;  
        Sat, 5 Jan 2008 09:14:15 -0500  
Received: FROM paploo.uhi.ac.uk (app1.prod.collab.uhi.ac.uk [194.35.219.  
184])  
        BY holes.mr.itd.umich.edu ID 477F90B0.2DB2F.12494 ;  
        5 Jan 2008 09:14:10 -0500  
Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])  
        by paploo.uhi.ac.uk (Postfix) with ESMTP id 5F919BC2F2;  
        Sat, 5 Jan 2008 14:14:05 -0000 (GMT)
```

```
In [15]: file_path="C:\\Users\\omkar\\OneDrive\\Documents\\Data science\\Naresh IT\\  
file=open(file_path,mode='r')  
print(file.read())
```

```
From stephen.marquard@uct.ac.za Sat Jan  5 09:14:16 2008  
Return-Path: <postmaster@collab.sakaiproject.org>  
Received: from murder (mail.umich.edu [141.211.14.90])  
        by frankenstein.mail.umich.edu (Cyrus v2.3.8) with LMTPA;  
        Sat, 05 Jan 2008 09:14:16 -0500  
X-Sieve: CMU Sieve 2.3  
Received: from murder ([unix socket])  
        by mail.umich.edu (Cyrus v2.2.12) with LMTPA;  
        Sat, 05 Jan 2008 09:14:16 -0500  
Received: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu [141.211.1  
4.79])  
        by flawless.mail.umich.edu () with ESMTP id m05EEFR1013674;  
        Sat, 5 Jan 2008 09:14:15 -0500  
Received: FROM paploo.uhi.ac.uk (app1.prod.collab.uhi.ac.uk [194.35.219.  
184])  
        BY holes.mr.itd.umich.edu ID 477F90B0.2DB2F.12494 ;  
        5 Jan 2008 09:14:10 -0500  
Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])  
        by paploo.uhi.ac.uk (Postfix) with ESMTP id 5F919BC2F2;  
        Sat, 5 Jan 2008 14:14:05 -0000 (GMT)
```

```
In [16]: file_path="C:\\Users\\omkar\\OneDrive\\Documents\\Data science\\Naresh IT\\Data science\\Batch-5_Dec"
file=open(file_path,
          mode='r',
          encoding='utf-8-sig')
print(file.read())
```

From stephen.marquard@uct.ac.za Sat Jan 5 09:14:16 2008
Return-Path: <postmaster@collab.sakaiproject.org>
Received: from murder (mail.umich.edu [141.211.14.90])
by frankenstein.mail.umich.edu (Cyrus v2.3.8) with LMTPA;
Sat, 05 Jan 2008 09:14:16 -0500
X-Sieve: CMU Sieve 2.3
Received: from murder ([unix socket])
by mail.umich.edu (Cyrus v2.2.12) with LMTPA;
Sat, 05 Jan 2008 09:14:16 -0500
Received: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu [141.211.14.79])
by flawless.mail.umich.edu () with ESMTP id m05EEFR1013674;
Sat, 5 Jan 2008 09:14:15 -0500
Received: FROM paploo.uhi.ac.uk (app1.prod.collab.uhi.ac.uk [194.35.219.184])
BY holes.mr.itd.umich.edu ID 477F90B0.2DB2F.12494 ;
5 Jan 2008 09:14:10 -0500
Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
by paploo.uhi.ac.uk (Postfix) with ESMTP id 5F919BC2F2;
Sat, 5 Jan 2008 09:14:10 -0500 (GMT)

- mode:
 - are you reading? 'r'
 - are you writing? 'w'
 - the default argument is read
- encode
 - spl characters : '»¿
 - encode= 'utf-8-sig'

```
In [ ]: file_path="C:\\Users\\omkar\\OneDrive\\Documents\\Data science\\Naresh IT\\Data science\\Batch-5_Dec"
file=open(file_path,
          mode='r',
          encoding='utf-8-sig')
print(file.read())
```

```
In [ ]: the full file path steps valid ,
when your data and python notebook both are in different places
```

```
In [1]: import os
os.getcwd()
```

```
Out[1]: 'C:\\Users\\omkar\\OneDrive\\Documents\\Data science\\Naresh IT\\Data science\\Batch-5_Dec'
```

```
In [ ]:
```

\$Method-2\$:

keep your python file and data file in same location

```
In [2]: file_path="mbox-short.txt"
file=open(file_path,
           mode='r',
           encoding='utf-8-sig')
print(file.read())
```

```
From stephen.marquard@uct.ac.za Sat Jan  5 09:14:16 2008
Return-Path: <postmaster@collab.sakaiproject.org>
Received: from murder (mail.umich.edu [141.211.14.90])
        by frankenstein.mail.umich.edu (Cyrus v2.3.8) with LMTPA;
        Sat, 05 Jan 2008 09:14:16 -0500
X-Sieve: CMU Sieve 2.3
Received: from murder ([unix socket])
        by mail.umich.edu (Cyrus v2.2.12) with LMTPA;
        Sat, 05 Jan 2008 09:14:16 -0500
Received: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu [141.211.1
4.79])
        by flawless.mail.umich.edu () with ESMTP id m05EEFR1013674;
        Sat, 5 Jan 2008 09:14:15 -0500
Received: FROM paploo.uhi.ac.uk (app1.prod.collab.uhi.ac.uk [194.35.219.
184])
        BY holes.mr.itd.umich.edu ID 477F90B0.2DB2F.12494 ;
        5 Jan 2008 09:14:10 -0500
Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
        by paploo.uhi.ac.uk (Postfix) with ESMTP id 5F919BC2F2;
        Sat, 5 Jan 2008 14:14:05 -0000 (GMT)
```

```
In [ ]: 'C:\\Users\\omkar\\OneDrive\\Documents\\Data science\\Naresh IT\\Data science\\Batch-5_Dec\\mbox-short.txt'
```

```
In [4]: os.getcwd()+"\\"+file_path
```

```
Out[4]: 'C:\\Users\\omkar\\OneDrive\\Documents\\Data science\\Naresh IT\\Data science\\Batch-5_Dec\\mbox-short.txt'
```

```
In [5]: import os
file_name="mbox-short.txt"
filepath=os.getcwd()+"\\"+file_name
file=open(file_path,
          mode='r',
          encoding='utf-8-sig')
print(file.read())
```

```
From stephen.marquard@uct.ac.za Sat Jan  5 09:14:16 2008
Return-Path: <postmaster@collab.sakaiproject.org>
Received: from murder (mail.umich.edu [141.211.14.90])
        by frankenstein.mail.umich.edu (Cyrus v2.3.8) with LMTPA;
        Sat, 05 Jan 2008 09:14:16 -0500
X-Sieve: CMU Sieve 2.3
Received: from murder ([unix socket])
        by mail.umich.edu (Cyrus v2.2.12) with LMTPA;
        Sat, 05 Jan 2008 09:14:16 -0500
Received: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu [141.211.1
4.79])
        by flawless.mail.umich.edu () with ESMTP id m05EEFR1013674;
        Sat, 5 Jan 2008 09:14:15 -0500
Received: FROM paploo.uhi.ac.uk (app1.prod.collab.uhi.ac.uk [194.35.219.
184])
        BY holes.mr.itd.umich.edu ID 477F90B0.2DB2F.12494 ;
        5 Jan 2008 09:14:10 -0500
Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
        by paploo.uhi.ac.uk (Postfix) with ESMTP id 5F919BC2F2;
```

\$Method-3\$

using with keyword

```
In [6]: file_path="mbox-short.txt"
file=open(file_path)
print(file.read())

file_path="mbox-short.txt"
with open(file_path) as file:
    print(file.read())
```

```
From stephen.marquard@uct.ac.za Sat Jan  5 09:14:16 2008
Return-Path: <postmaster@collab.sakaiproject.org>
Received: from murder (mail.umich.edu [141.211.14.90])
        by frankenstein.mail.umich.edu (Cyrus v2.3.8) with LMTPA;
        Sat, 05 Jan 2008 09:14:16 -0500
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Received: from murder ([unix socket])
        by mail.umich.edu (Cyrus v2.2.12) with LMTPA;
        Sat, 05 Jan 2008 09:14:16 -0500
Received: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu [141.211.1
4.79])
        by flawless.mail.umich.edu () with ESMTP id m05EEFR1013674;
        Sat, 5 Jan 2008 09:14:15 -0500
Received: FROM paploo.uhi.ac.uk (app1.prod.collab.uhi.ac.uk [194.35.219.
184])
        BY holes.mr.itd.umich.edu ID 477F90B0.2DB2F.12494 ;
        5 Jan 2008 09:14:10 -0500
Received: from paploo.uhi.ac.uk (localhost [127.0.0.1])
        by paploo.uhi.ac.uk (Postfix) with ESMTP id 5F919BC2F2;
```

```
In [ ]: #100file_path="mbox-short.txt"
#file=open(file_path)
#print(file.read())
```

- file session will close
- to avoid this will go with keyword

```
In [ ]: ##### Method-1 #####
# Python file and data file in different location
file_path="C:\\Users\\omkar\\OneDrive\\Documents\\Data science\\Naresh IT\\I
file=open(file_path,
          mode='r',
          encoding='utf-8-sig')
print(file.read())

##### Method-2 #####
# Python file and data file in same location
file_path="mbox-short.txt"
file=open(file_path,
          mode='r',
          encoding='utf-8-sig')
print(file.read())

##### Method-3#####
# With keyword avoid the closing of working session
file_path="mbox-short.txt"
with open(file_path) as file:
    print(file.read())
```

```
In [15]: file_path="mbox-short.txt"
file=open(file_path)
for line in file: # Line by line we are reading the file
    print(line)
```

From stephen.marquard@uct.ac.za Sat Jan 5 09:14:16 2008

Return-Path: <postmaster@collab.sakaiproject.org>

Received: from murder (mail.umich.edu [141.211.14.90])

by frankenstein.mail.umich.edu (Cyrus v2.3.8) with LMTPA;

Sat, 05 Jan 2008 09:14:16 -0500

X-Sieve: CMU Sieve 2.3

Received: from murder ([unix socket])

by mail.umich.edu (Cyrus v2.2.12) with LMTPA;

Sat, 05 Jan 2008 09:14:16 -0500

Received: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu [141.211.1
4.701])

```
In [17]: #Q1) write a program count the number of lines in a given file
file_path="mbox-short.txt"
file=open(file_path)
count=0
for line in file:    # Line by line we are reading the file
    count=count+1
print(count)
```

1910

```
In [ ]: # type(file)
# _io.TextIOWrapper

# for i in file:
#     print(i)
# -----
# ValueError                                Traceback (most recent call last)
# /var/folders/_5/f439c61s71nc6hfdz2f2ndtr0000gn/T/ipykernel_63911/50471970
# ----> 1 for i in file:
#       2     print(i)

# ValueError: I/O operation on closed file.
```

\$enumerate\$

- it will give two outputs
- index of line
- content

```
In [23]: file_path="mbox-short.txt"
file=open(file_path)
for index,line in enumerate(file):
    print(index,line)
```

```
0 From stephen.marquard@uct.ac.za Sat Jan  5 09:14:16 2008
1 Return-Path: <postmaster@collab.sakaiproject.org>
2 Received: from murder (mail.umich.edu [141.211.14.90])
3     by frankenstein.mail.umich.edu (Cyrus v2.3.8) with LMTPA;
4     Sat, 05 Jan 2008 09:14:16 -0500
5 X-Sieve: CMU Sieve 2.3
6 Received: from murder ([unix socket])
7     by mail.umich.edu (Cyrus v2.2.12) with LMTPA;
8     Sat, 05 Jan 2008 09:14:16 -0500
9 Received: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu [141.21
1 14 2011)
```


In [35]: *#Q2) Count the number of lines by using enumerate index
dont use counter*

```
file_path="mbox-short.txt"
file=open(file_path)
for index,_ in enumerate(file):
    continue # pass
index+1

# enumerate : 2
# if you provide one variable=(a,b)
# but you dont want use line= _
# _ communicating
```

Out[35]: 1910

In [38]:

```
file_path="mbox-short.txt"
file=open(file_path)
l1=[index for index,_ in enumerate(file)]
len(l1)
```

Out[38]: 1910

In []: From stephen.marquard@uct.ac.za Sat Jan 5 09:14:16 2008

Return-Path: <postmaster@collab.sakaiproject.org>

Received: from murder (mail.umich.edu [141.211.14.90])

by frankenstein.mail.umich.edu (Cyrus v2.3.8) with LMTPA;

Sat, 05 Jan 2008 09:14:16 -0500

X-Sieve: CMU Sieve 2.3

In [42]: *#Q3) make all the lines in align*

```
file_path="mbox-short.txt"
file=open(file_path)
for line in file:
    print(line.lstrip())
```

From stephen.marquard@uct.ac.za Sat Jan 5 09:14:16 2008

Return-Path: <postmaster@collab.sakaiproject.org>

Received: from murder (mail.umich.edu [141.211.14.90])

by frankenstein.mail.umich.edu (Cyrus v2.3.8) with LMTPA;

Sat, 05 Jan 2008 09:14:16 -0500

X-Sieve: CMU Sieve 2.3

Received: from murder ([unix socket])

by mail.umich.edu (Cyrus v2.2.12) with LMTPA;

Sat, 05 Jan 2008 09:14:16 -0500

Received: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu [141.211.14.79])

In [46]: *#Q4) print first lines*

```
filename="mbox-short.txt"
count=0
with open(filename) as file:
    for index,line in enumerate(file):
        print(index,line)
        if index==10:
            break
```

0 From stephen.marquard@uct.ac.za Sat Jan 5 09:14:16 2008

1 Return-Path: <postmaster@collab.sakaiproject.org>

2 Received: from murder (mail.umich.edu [141.211.14.90])

3 by frankenstein.mail.umich.edu (Cyrus v2.3.8) with LMTPA;

4 Sat, 05 Jan 2008 09:14:16 -0500

5 X-Sieve: CMU Sieve 2.3

6 Received: from murder ([unix socket])

7 by mail.umich.edu (Cyrus v2.2.12) with LMTPA;

8 Sat, 05 Jan 2008 09:14:16 -0500

9 Received: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu [141.211.14.79])

10 by flawless.mail.umich.edu () with ESMTP id m05EEFR1013674;

```
In [47]: file_path="mbox-short.txt"
file=open(file_path)
for index,line in enumerate(file):
    if index<=9:
        print(line)

for index,line in enumerate(file):
    print(line)
    if index==10:
        break
```

From stephen.marquard@uct.ac.za Sat Jan 5 09:14:16 2008

Return-Path: <postmaster@collab.sakaiproject.org>

Received: from murder (mail.umich.edu [141.211.14.90])

by frankenstein.mail.umich.edu (Cyrus v2.3.8) with LMTPA;

Sat, 05 Jan 2008 09:14:16 -0500

X-Sieve: CMU Sieve 2.3

Received: from murder ([unix socket])

by mail.umich.edu (Cyrus v2.2.12) with LMTPA;

Sat, 05 Jan 2008 09:14:16 -0500

Received: from holes.mr.itd.umich.edu (holes.mr.itd.umich.edu [141.211.14.79])

```
In [ ]: #Q1) write a program count the number of lines in a given file
#Q2) Count the number of lines by using enumerate index
# dont use counter
#Q3) make all the lines in align
#Q4) print first lines
```

Q)5 get the lines which are starting with From, save that in a list

```
In [7]: file_path="mbox-short.txt"
file=open(file_path)

##### M-1#####
lst=[]
for line in file:
    if line.startswith("From"):
        lst.append(line)
lst
##### M-2#####
```

```
Out[7]: ['From stephen.marquard@uct.ac.za Sat Jan  5 09:14:16 2008\n',
'From: stephen.marquard@uct.ac.za\n',
'From louis@media.berkeley.edu Fri Jan  4 18:10:48 2008\n',
'From: louis@media.berkeley.edu\n',
'From zqian@umich.edu Fri Jan  4 16:10:39 2008\n',
'From: zqian@umich.edu\n',
'From rjlowe@iupui.edu Fri Jan  4 15:46:24 2008\n',
'From: rjlowe@iupui.edu\n',
'From zqian@umich.edu Fri Jan  4 15:03:18 2008\n',
'From: zqian@umich.edu\n',
'From rjlowe@iupui.edu Fri Jan  4 14:50:18 2008\n',
'From: rjlowe@iupui.edu\n',
'From cwen@iupui.edu Fri Jan  4 11:37:30 2008\n',
'From: cwen@iupui.edu\n',
'From cwen@iupui.edu Fri Jan  4 11:35:08 2008\n',
'From: cwen@iupui.edu\n',
'From gsilver@umich.edu Fri Jan  4 11:12:37 2008\n',
'From: gsilver@umich.edu\n',
'From gsilver@umich.edu Fri Jan  4 11:11:52 2008\n',
'From: gsilver@umich.edu\n',
'From zqian@umich.edu Fri Jan  4 11:11:03 2008\n',
'From: zqian@umich.edu\n',
'From gsilver@umich.edu Fri Jan  4 11:10:22 2008\n',
'From: gsilver@umich.edu\n',
'From wagnermr@iupui.edu Fri Jan  4 10:38:42 2008\n',
'From: wagnermr@iupui.edu\n',
'From zqian@umich.edu Fri Jan  4 10:17:43 2008\n',
'From: zqian@umich.edu\n',
'From antranig@caret.cam.ac.uk Fri Jan  4 10:04:14 2008\n',
'From: antranig@caret.cam.ac.uk\n',
'From gopal.ramasammycook@gmail.com Fri Jan  4 09:05:31 2008\n',
'From: gopal.ramasammycook@gmail.com\n',
'From david.horwitz@uct.ac.za Fri Jan  4 07:02:32 2008\n',
'From: david.horwitz@uct.ac.za\n',
'From david.horwitz@uct.ac.za Fri Jan  4 06:08:27 2008\n',
'From: david.horwitz@uct.ac.za\n',
'From david.horwitz@uct.ac.za Fri Jan  4 04:49:08 2008\n',
'From: david.horwitz@uct.ac.za\n',
'From david.horwitz@uct.ac.za Fri Jan  4 04:33:44 2008\n',
'From: david.horwitz@uct.ac.za\n',
'From stephen.marquard@uct.ac.za Fri Jan  4 04:07:34 2008\n',
'From: stephen.marquard@uct.ac.za\n',
'From louis@media.berkeley.edu Thu Jan  3 19:51:21 2008\n',
'From: louis@media.berkeley.edu\n',
'From louis@media.berkeley.edu Thu Jan  3 17:18:23 2008\n',
'From: louis@media.berkeley.edu\n',
'From ray@media.berkeley.edu Thu Jan  3 17:07:00 2008\n',
'From: ray@media.berkeley.edu\n',
'From cwen@iupui.edu Thu Jan  3 16:34:40 2008\n',
'From: cwen@iupui.edu\n',
'From cwen@iupui.edu Thu Jan  3 16:29:07 2008\n',
'From: cwen@iupui.edu\n',
'From cwen@iupui.edu Thu Jan  3 16:23:48 2008\n',
'From: cwen@iupui.edu\n']
```

```
In [8]: ##### M-2#####
file_path="mbox-short.txt"
with open(file_path) as file:
    lst=[line for line in file if line.startswith("From")]
lst
```

```
Out[8]: ['From stephen.marquard@uct.ac.za Sat Jan  5 09:14:16 2008\n',
'From: stephen.marquard@uct.ac.za\n',
'From louis@media.berkeley.edu Fri Jan  4 18:10:48 2008\n',
'From: louis@media.berkeley.edu\n',
'From zqian@umich.edu Fri Jan  4 16:10:39 2008\n',
'From: zqian@umich.edu\n',
'From rjlowe@iupui.edu Fri Jan  4 15:46:24 2008\n',
'From: rjlowe@iupui.edu\n',
'From zqian@umich.edu Fri Jan  4 15:03:18 2008\n',
'From: zqian@umich.edu\n',
'From rjlowe@iupui.edu Fri Jan  4 14:50:18 2008\n',
'From: rjlowe@iupui.edu\n',
'From cwen@iupui.edu Fri Jan  4 11:37:30 2008\n',
'From: cwen@iupui.edu\n',
'From cwen@iupui.edu Fri Jan  4 11:35:08 2008\n',
'From: cwen@iupui.edu\n',
'From gsilver@umich.edu Fri Jan  4 11:12:37 2008\n',
'From: gsilver@umich.edu\n',
'From gsilver@umich.edu Fri Jan  4 11:11:52 2008\n',
'From: gsilver@umich.edu\n',
'From zqian@umich.edu Fri Jan  4 11:11:03 2008\n',
'From: zqian@umich.edu\n',
'From gsilver@umich.edu Fri Jan  4 11:10:22 2008\n',
'From: gsilver@umich.edu\n',
'From wagnermr@iupui.edu Fri Jan  4 10:38:42 2008\n',
'From: wagnermr@iupui.edu\n',
'From zqian@umich.edu Fri Jan  4 10:17:43 2008\n',
'From: zqian@umich.edu\n',
'From antranig@caret.cam.ac.uk Fri Jan  4 10:04:14 2008\n',
'From: antranig@caret.cam.ac.uk\n',
'From gopal.ramasammycook@gmail.com Fri Jan  4 09:05:31 2008\n',
'From: gopal.ramasammycook@gmail.com\n',
'From david.horwitz@uct.ac.za Fri Jan  4 07:02:32 2008\n',
'From: david.horwitz@uct.ac.za\n',
'From david.horwitz@uct.ac.za Fri Jan  4 06:08:27 2008\n',
'From: david.horwitz@uct.ac.za\n',
'From david.horwitz@uct.ac.za Fri Jan  4 04:49:08 2008\n',
'From: david.horwitz@uct.ac.za\n',
'From david.horwitz@uct.ac.za Fri Jan  4 04:33:44 2008\n',
'From: david.horwitz@uct.ac.za\n',
'From stephen.marquard@uct.ac.za Fri Jan  4 04:07:34 2008\n',
'From: stephen.marquard@uct.ac.za\n',
'From louis@media.berkeley.edu Thu Jan  3 19:51:21 2008\n',
'From: louis@media.berkeley.edu\n',
'From louis@media.berkeley.edu Thu Jan  3 17:18:23 2008\n',
'From: louis@media.berkeley.edu\n',
'From ray@media.berkeley.edu Thu Jan  3 17:07:00 2008\n',
'From: ray@media.berkeley.edu\n',
'From cwen@iupui.edu Thu Jan  3 16:34:40 2008\n',
'From: cwen@iupui.edu\n',
'From cwen@iupui.edu Thu Jan  3 16:29:07 2008\n',
'From: cwen@iupui.edu\n',
'From cwen@iupui.edu Thu Jan  3 16:23:48 2008\n',
'From: cwen@iupui.edu\n']
```

Q6) Remove the \n in the above output

```
In [17]: file_path="mbox-short.txt"
file=open(file_path)
lst=[line for line in file if line.startswith('From')]
lst_n=[line.rstrip() for line in lst]
len(lst_n)
```

Out[17]: 54

Q7) get the same output shown in (Q6). But with out taking the help of (Q5) outpt

```
In [20]: file_path = "mbox-short.txt"
file = open(file_path)
lst=[line.rstrip('\n') for line in file if line.startswith("From")]
lst
```

```
Out[20]: ['From stephen.marquard@uct.ac.za Sat Jan  5 09:14:16 2008',
'From: stephen.marquard@uct.ac.za',
'From louis@media.berkeley.edu Fri Jan  4 18:10:48 2008',
'From: louis@media.berkeley.edu',
'From zqian@umich.edu Fri Jan  4 16:10:39 2008',
'From: zqian@umich.edu',
'From rjlowe@iupui.edu Fri Jan  4 15:46:24 2008',
'From: rjlowe@iupui.edu',
'From zqian@umich.edu Fri Jan  4 15:03:18 2008',
'From: zqian@umich.edu',
'From rjlowe@iupui.edu Fri Jan  4 14:50:18 2008',
'From: rjlowe@iupui.edu',
'From cwen@iupui.edu Fri Jan  4 11:37:30 2008',
'From: cwen@iupui.edu',
'From cwen@iupui.edu Fri Jan  4 11:35:08 2008',
'From: cwen@iupui.edu',
'From gsilver@umich.edu Fri Jan  4 11:12:37 2008',
'From: gsilver@umich.edu',
'From gsilver@umich.edu Fri Jan  4 11:11:52 2008',
'From: gsilver@umich.edu',
'From zqian@umich.edu Fri Jan  4 11:11:03 2008',
'From: zqian@umich.edu',
'From gsilver@umich.edu Fri Jan  4 11:10:22 2008',
'From: gsilver@umich.edu',
'From wagnermr@iupui.edu Fri Jan  4 10:38:42 2008',
'From: wagnermr@iupui.edu',
'From zqian@umich.edu Fri Jan  4 10:17:43 2008',
'From: zqian@umich.edu',
'From antranig@caret.cam.ac.uk Fri Jan  4 10:04:14 2008',
'From: antranig@caret.cam.ac.uk',
'From gopal.ramasammycook@gmail.com Fri Jan  4 09:05:31 2008',
'From: gopal.ramasammycook@gmail.com',
'From david.horwitz@uct.ac.za Fri Jan  4 07:02:32 2008',
'From: david.horwitz@uct.ac.za',
'From david.horwitz@uct.ac.za Fri Jan  4 06:08:27 2008',
'From: david.horwitz@uct.ac.za',
'From david.horwitz@uct.ac.za Fri Jan  4 04:49:08 2008',
'From: david.horwitz@uct.ac.za',
'From david.horwitz@uct.ac.za Fri Jan  4 04:33:44 2008',
'From: david.horwitz@uct.ac.za',
'From stephen.marquard@uct.ac.za Fri Jan  4 04:07:34 2008',
'From: stephen.marquard@uct.ac.za',
'From louis@media.berkeley.edu Thu Jan  3 19:51:21 2008',
'From: louis@media.berkeley.edu',
'From louis@media.berkeley.edu Thu Jan  3 17:18:23 2008',
'From: louis@media.berkeley.edu',
'From ray@media.berkeley.edu Thu Jan  3 17:07:00 2008',
'From: ray@media.berkeley.edu',
'From cwen@iupui.edu Thu Jan  3 16:34:40 2008',
'From: cwen@iupui.edu',
'From cwen@iupui.edu Thu Jan  3 16:29:07 2008',
'From: cwen@iupui.edu',
'From cwen@iupui.edu Thu Jan  3 16:23:48 2008',
'From: cwen@iupui.edu']
```


Q8) Extract email ids from (Q7) output

```
In [21]: file_path = "mbox-short.txt"
#file=open(file_path)
with open(file_path) as file:
    lst=[line.rstrip() for line in file if line.startswith('From')]
    lst_emails=[email.split()[1] for email in lst ]
    lst_emails
```

```
Out[21]: ['stephen.marquard@uct.ac.za',
'stephen.marquard@uct.ac.za',
'louis@media.berkeley.edu',
'louis@media.berkeley.edu',
'zqian@umich.edu',
'zqian@umich.edu',
'rjlowe@iupui.edu',
'rjlowe@iupui.edu',
'zqian@umich.edu',
'zqian@umich.edu',
'rjlowe@iupui.edu',
'rjlowe@iupui.edu',
'cwen@iupui.edu',
'cwen@iupui.edu',
'cwen@iupui.edu',
'cwen@iupui.edu',
'gsilver@umich.edu',
'gsilver@umich.edu',
'gsilver@umich.edu',
'gsilver@umich.edu',
'zqian@umich.edu',
'zqian@umich.edu',
'gsilver@umich.edu',
'gsilver@umich.edu',
'wagnermr@iupui.edu',
'wagnermr@iupui.edu',
'zqian@umich.edu',
'zqian@umich.edu',
'antranig@caret.cam.ac.uk',
'antranig@caret.cam.ac.uk',
'gopal.ramasammycook@gmail.com',
'gopal.ramasammycook@gmail.com',
'david.horwitz@uct.ac.za',
'david.horwitz@uct.ac.za',
'david.horwitz@uct.ac.za',
'david.horwitz@uct.ac.za',
'david.horwitz@uct.ac.za',
'david.horwitz@uct.ac.za',
'david.horwitz@uct.ac.za',
'david.horwitz@uct.ac.za',
'stephen.marquard@uct.ac.za',
'stephen.marquard@uct.ac.za',
'louis@media.berkeley.edu',
'louis@media.berkeley.edu',
'louis@media.berkeley.edu',
'louis@media.berkeley.edu',
'ray@media.berkeley.edu',
'ray@media.berkeley.edu',
'cwen@iupui.edu',
'cwen@iupui.edu',
'cwen@iupui.edu',
'cwen@iupui.edu',
'cwen@iupui.edu',
'cwen@iupui.edu']
```

```
In [22]: file_name = "mbox-short.txt"
file = open(file_name)
f_list = [line.rstrip() for line in file if line.startswith('From')]

for line in f_list:
    line_list = [line for line in line.split() if '@' in line]
    print(line_list)
```

['stephen.marquard@uct.ac.za']
['stephen.marquard@uct.ac.za']
['louis@media.berkeley.edu']
['louis@media.berkeley.edu']
['zqian@umich.edu']
['zqian@umich.edu']
['rjlowe@iupui.edu']
['rjlowe@iupui.edu']
['zqian@umich.edu']
['zqian@umich.edu']
['rjlowe@iupui.edu']
['rjlowe@iupui.edu']
['cwen@iupui.edu']
['cwen@iupui.edu']
['cwen@iupui.edu']
['cwen@iupui.edu']
['gsilver@umich.edu']
['gsilver@umich.edu']
['gsilver@umich.edu']
['gsilver@umich.edu']
['zqian@umich.edu']
['zqian@umich.edu']
['gsilver@umich.edu']
['gsilver@umich.edu']
['wagnermr@iupui.edu']
['wagnermr@iupui.edu']
['zqian@umich.edu']
['zqian@umich.edu']
['antranig@caret.cam.ac.uk']
['antranig@caret.cam.ac.uk']
['gopal.ramasammycook@gmail.com']
['gopal.ramasammycook@gmail.com']
['david.horwitz@uct.ac.za']
['david.horwitz@uct.ac.za']
['david.horwitz@uct.ac.za']
['david.horwitz@uct.ac.za']
['david.horwitz@uct.ac.za']
['david.horwitz@uct.ac.za']
['david.horwitz@uct.ac.za']
['david.horwitz@uct.ac.za']
['stephen.marquard@uct.ac.za']
['stephen.marquard@uct.ac.za']
['louis@media.berkeley.edu']
['louis@media.berkeley.edu']
['louis@media.berkeley.edu']
['louis@media.berkeley.edu']
['ray@media.berkeley.edu']
['ray@media.berkeley.edu']
['cwen@iupui.edu']
['cwen@iupui.edu']
['cwen@iupui.edu']
['cwen@iupui.edu']
['cwen@iupui.edu']
['cwen@iupui.edu']

```
In [41]: file_path = "mbox-short.txt"
file = open(file_path)
lst=[line for line in file if line.startswith("From")]
lst_n=[line.rstrip() for line in lst if line.startswith("From")]
email=[line.split()[1] for line in lst_n]
lst[0],lst_n[0],email[0]

[line.rstrip().split()[1] for line in file if line.startswith("From")]
```

```
Out[41]: ('From stephen.marquard@uct.ac.za Sat Jan  5 09:14:16 2008\n',
'From stephen.marquard@uct.ac.za Sat Jan  5 09:14:16 2008',
'stephen.marquard@uct.ac.za')
```

Q9) Extract unique email ids with out using set operations

```
In [51]: file_path = "mbox-short.txt"
file = open(file_path)
emails=[line.rstrip().split()[1] for line in file if line.startswith("From")]

unique_emails=[]
for i in emails:
    if i not in unique_emails:
        unique_emails.append(i)

unique_emails
```

```
Out[51]: ['stephen.marquard@uct.ac.za',
'louis@media.berkeley.edu',
'zqian@umich.edu',
'rjlowe@iupui.edu',
'cwen@iupui.edu',
'gsilver@umich.edu',
'wagnermr@iupui.edu',
'antranig@caret.cam.ac.uk',
'gopal.ramasammycook@gmail.com',
'david.horwitz@uct.ac.za',
'ray@media.berkeley.edu']
```

```
In [ ]: sur_name=['Nallagoni','']
name=['Omkar']
email=['omkar.nallagoni@accenture.com']
company=['accenture']

EOD
```

```
In [ ]: pandas eda
```

```
In [ ]: file_path="mbox-short.txt"
with open(file_path,mode='r') as file:
    print(file.read())

# empty book: dummy.txt
# == > open 'w' == > .write("hello")
```

```
In [53]: # Different folder location
file_path="C:\\Users\\omkar\\OneDrive\\Documents\\dummy.txt"
with open(file_path,mode='w') as file:
    print(file.write("hello how are you"))
```

17

```
In [55]: # folder same as where python exist
file_path="dummy2.txt"
with open(file_path,mode='w',encoding='utf-8-sig') as file:
    print(file.write("hello how are you sanjay"))
```

24


```

In [ ]: ##### Part-1 #####
- Basic syntax

- print statements

- variables

- eval and input

- packages

- Conditional statements

- random math time

- try-except

- functions

- for

- while

##### Part-2#####
- Strings

- List

- Dictionary

- tuple and set

- file handling

- lambda

##### Part-3 (ML with python) #####
- Python OOPS

- class self init

- own packages

Statistics : 10 Sessions
EDA python: Pandas numpy matplotlib + 10 sessions

Jan 30 to Feb 15 : GitHub
Feb 15 to ML
March to DL: azure apps creation postman testing

vscode
Colab
GitHub
databricks

april end

monday to friday
saturday

```

- class notes
- text book
- assignments
- hacker rank
- 140 python
- great learning
- 1month
- python internship

```
In [57]: def addition():  
          num1=input() # string request.args.get()  
          num2=input() # string  
          return(int(num1)+int(num2))
```

```
addition()
```

```
10
```

```
20
```

Out[57]: 30

In []:

In []:

In []:

In []: